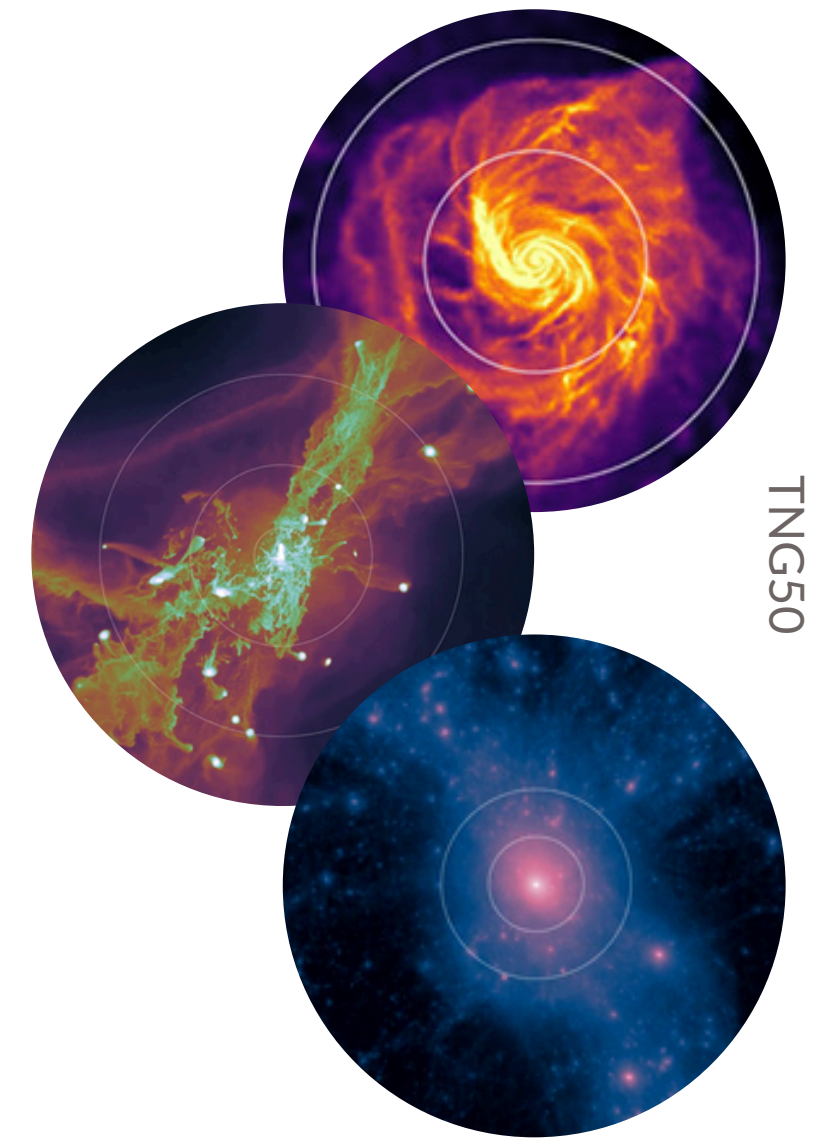


Spiral-Arm Morphology in IllustrisTNG50 Disk Galaxies and Its Relation to Dark Matter Halo Properties

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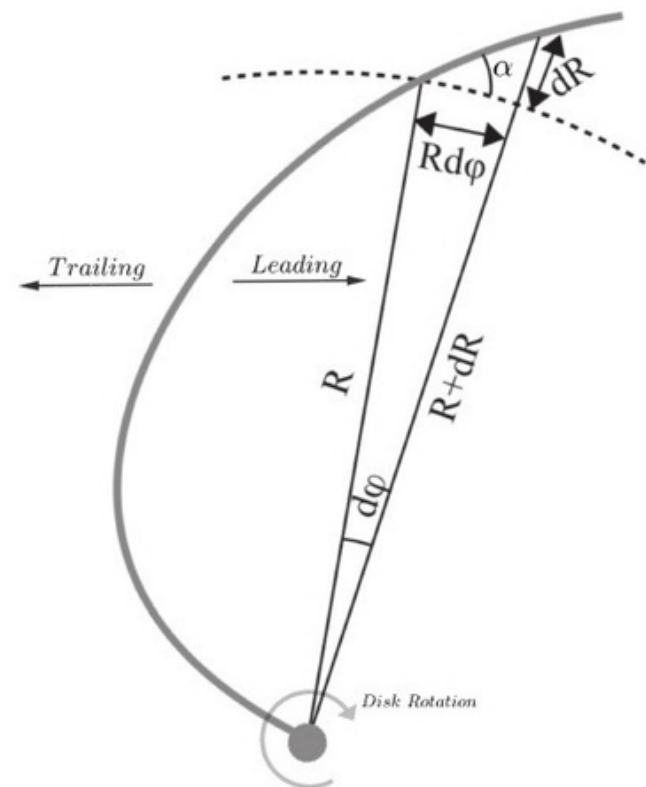


Spiral Arms in Disk Galaxies

Spiral-arm styles (Elmegreen et al., 1989; Yu & Ho, 2020):

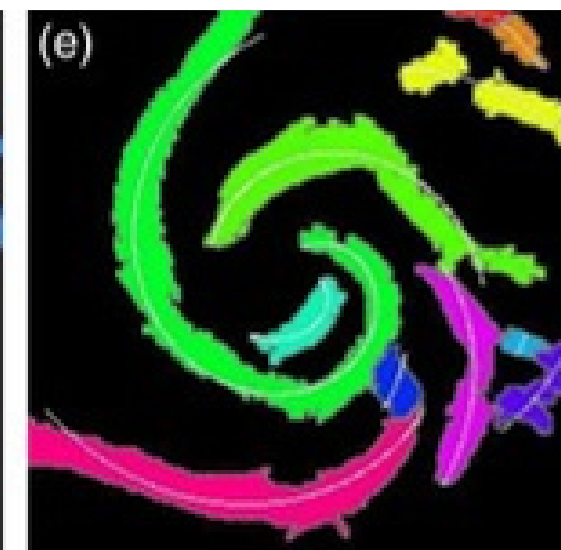
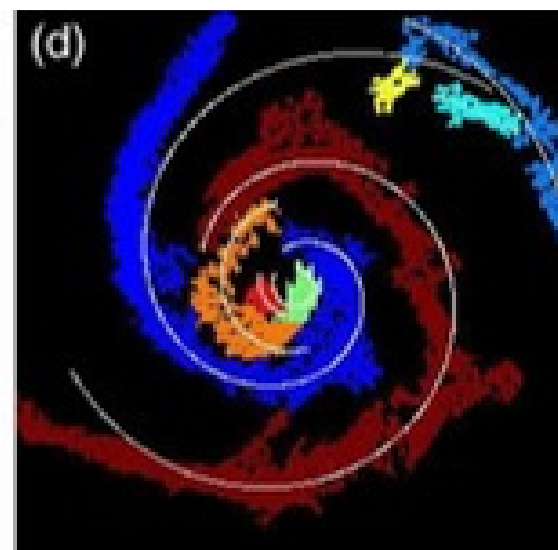
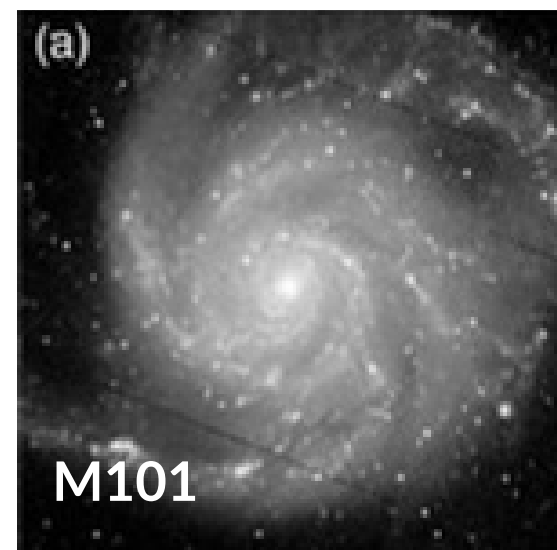
- ~10% grand design
- ~60% multi-armed
- ~30% flocculent

What are they? How do they form and move?



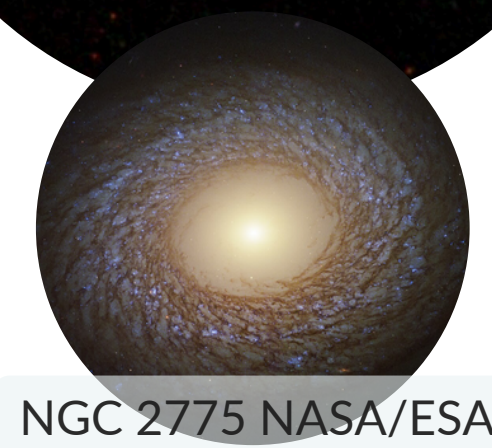
Pitch-Angle Definition

Observation



(Davis & Hayes, 2014)

Defining an 'Arm' Is Difficult



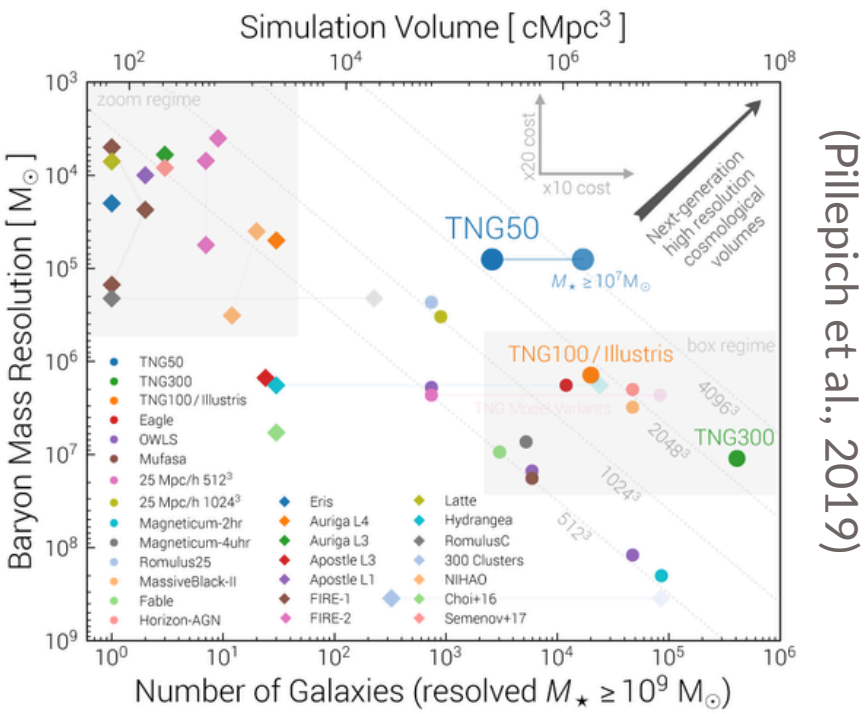
This context leads to a central question:

How do the properties of a galaxy's host dark matter halo relate to the morphology of its spiral arms?

- New patterns, new tests: disk–halo correlations can distinguish among density-wave, swing-amplification, and interaction scenarios.

What Sample Did We Use?

- TNG50
- 2.7 TB snapshot at $z = 0$
- Gas mass $> 10^{9.6} M_{\odot}$ (preselection)
- 377 subhalos total; 309 passed the filters



Parámetro	Variable
Coordenadas	x, y, z
Velocidad	vx, vy, vz
Momento angular	lxvel, lyvel, lzvel
Potencial gravitacional	Potential
Energía interna	U
Densidad	rho

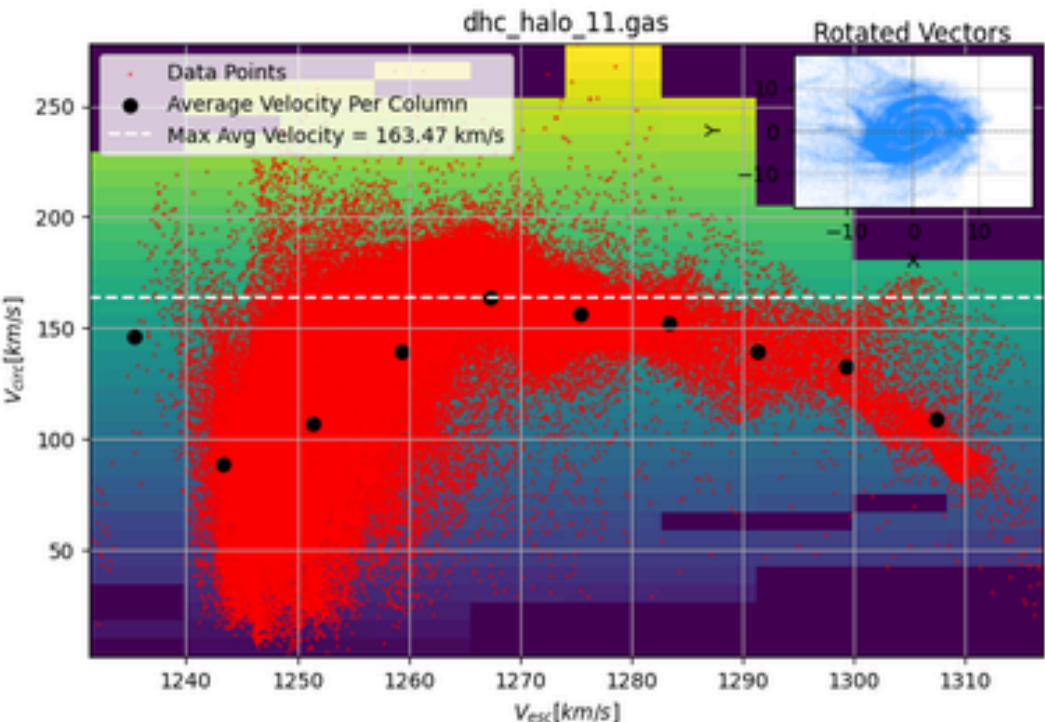
Table 1: Physical properties from IllustrisTNG gas cells (PartType0).

Parámetro	Variable
Masa total	logmass
Masa de gas	logmass0
Masa de materia oscura	logmass1
Masa estelar	logmass4
Radio	SubhaloHalfmassRad
Velocidad máxima	SubhaloVmax
Radio ($V_{circMax}$)	SubhaloVmaxRad

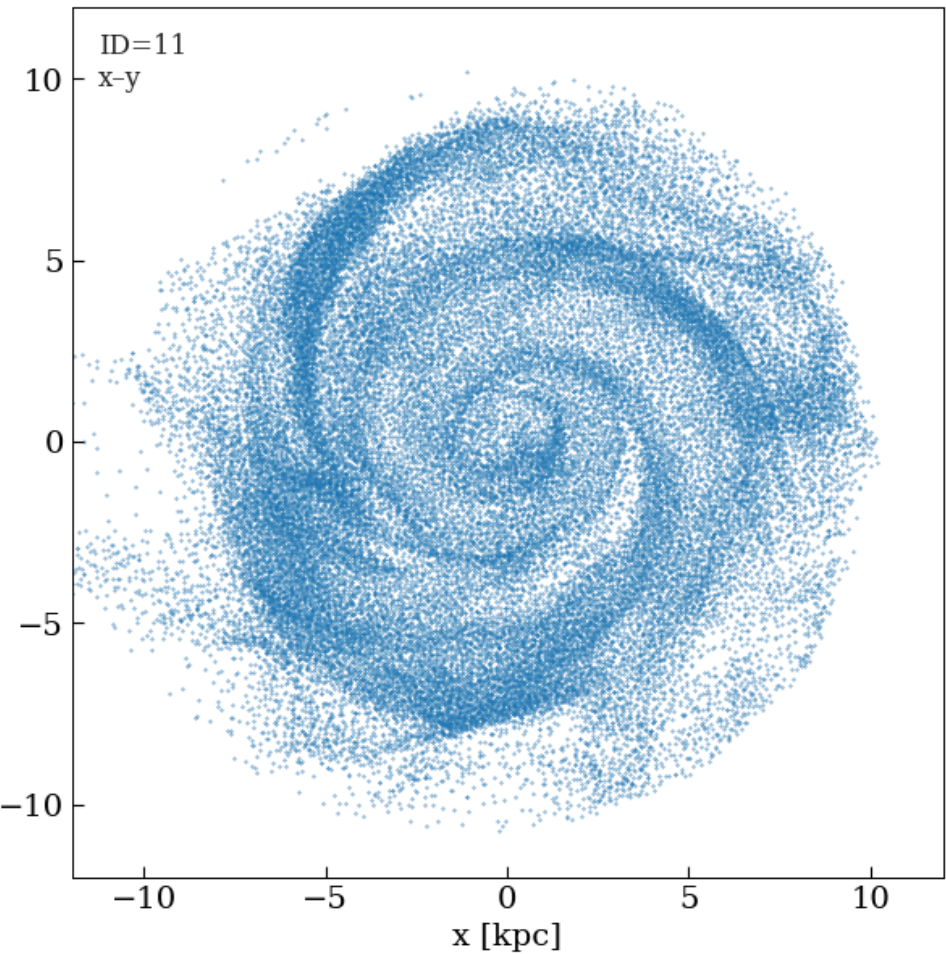
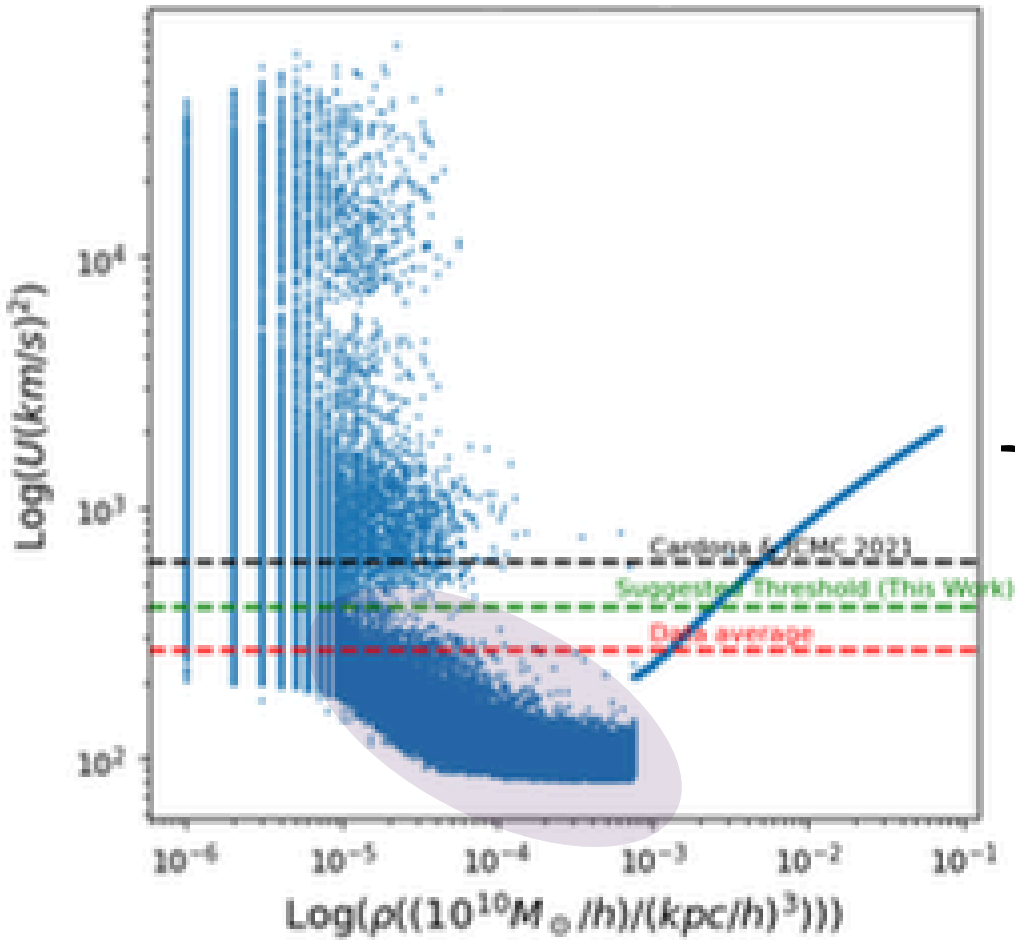
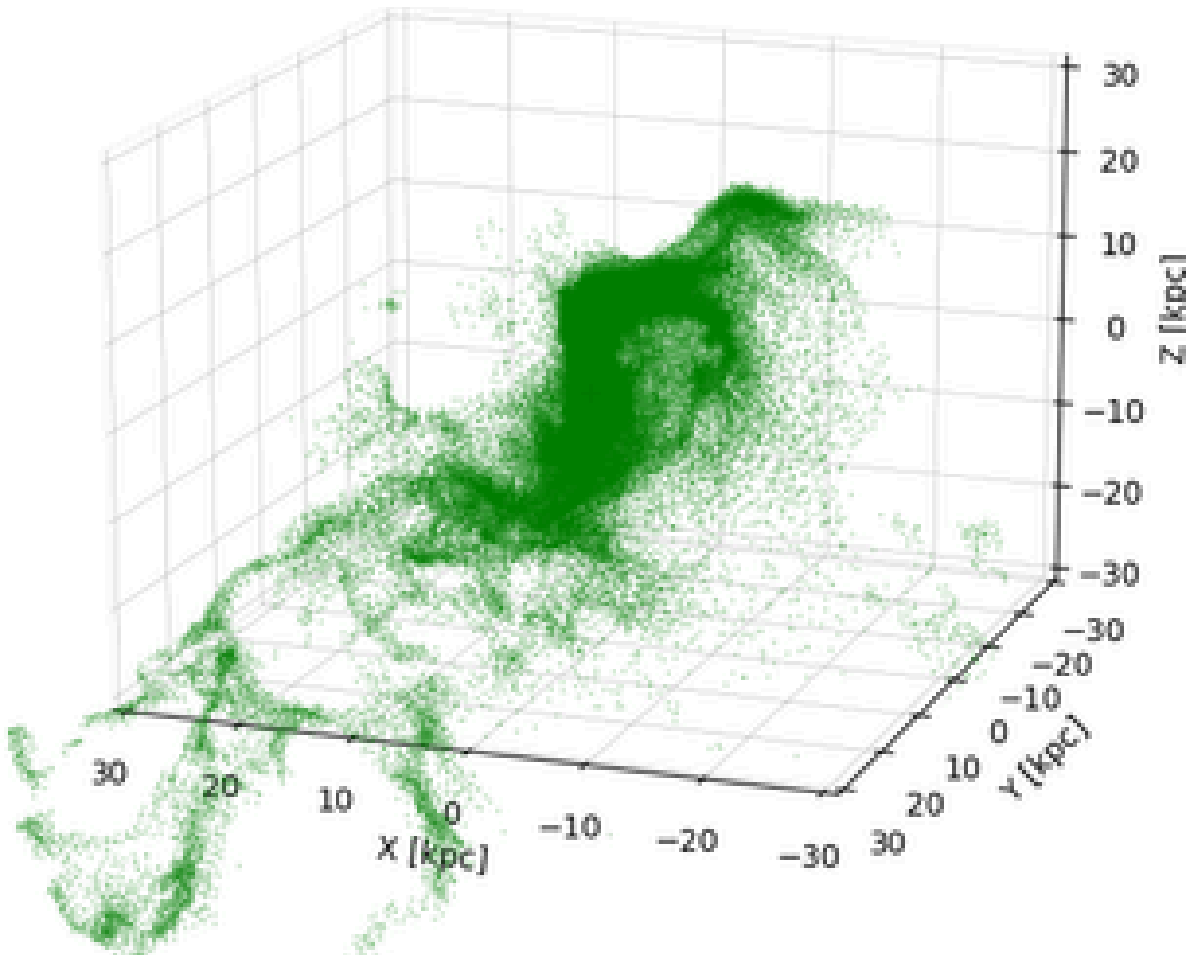
Table 2: Physical properties extracted for the subhalos.

Disk-Galaxy Selection

- ✓ Gas-Disk Inspection
 - ✓ Density-Based Spatial Clustering (DBSCAN)
 - ✓ Internal-Energy Selection to Isolate Disk Gas
 - ✓ Orbital-Circularity Selection
 - ✓ Face-On Deprojection
- 309 disks retained



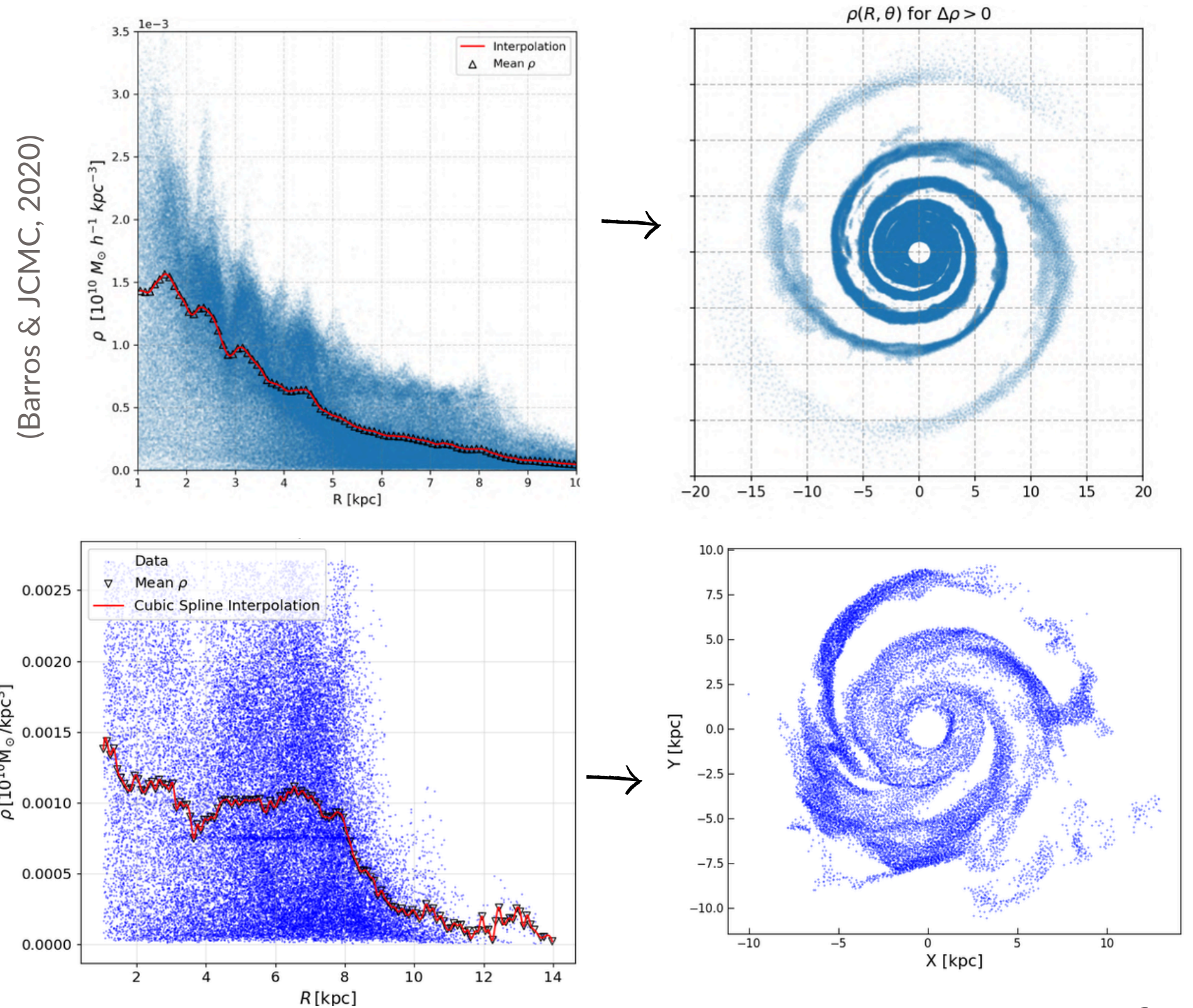
(Cardona & Muñoz-Cuartas, 2021)



Morphology Measurements

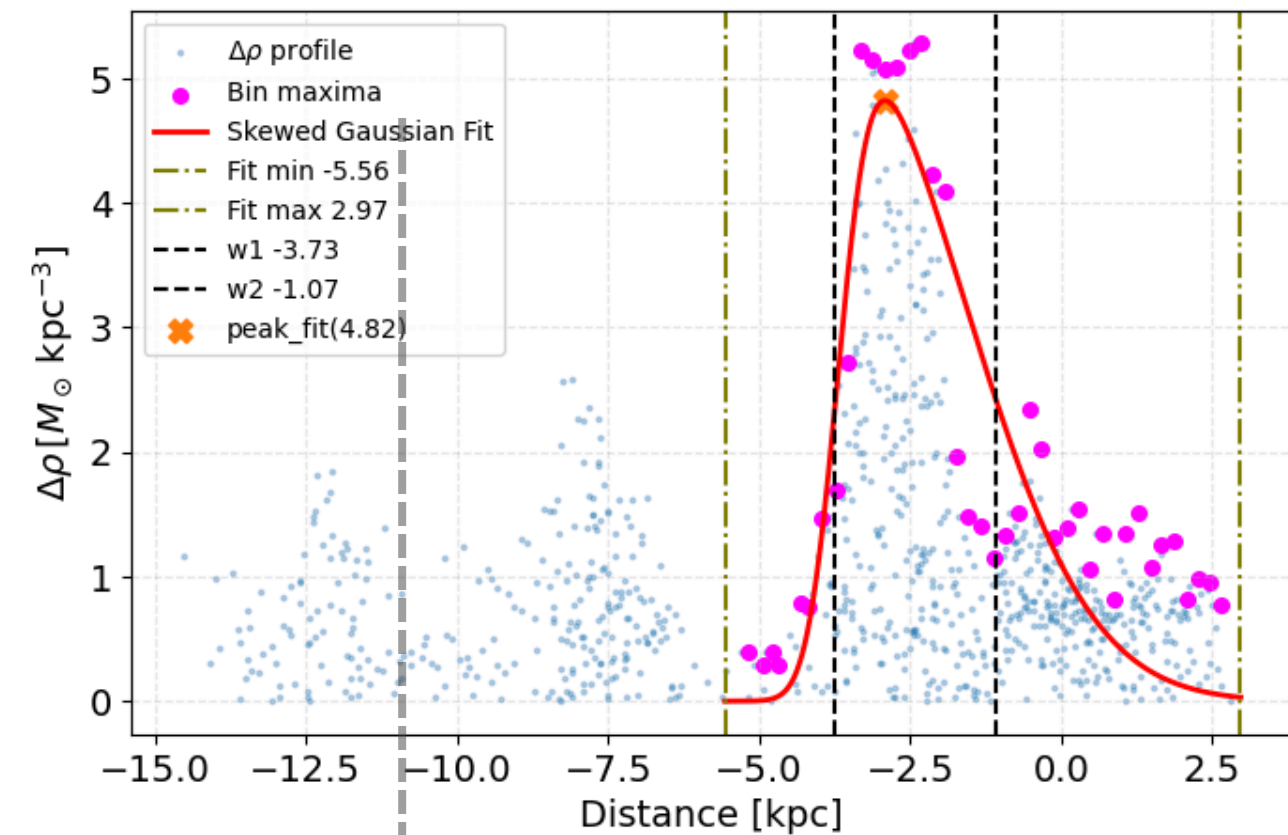
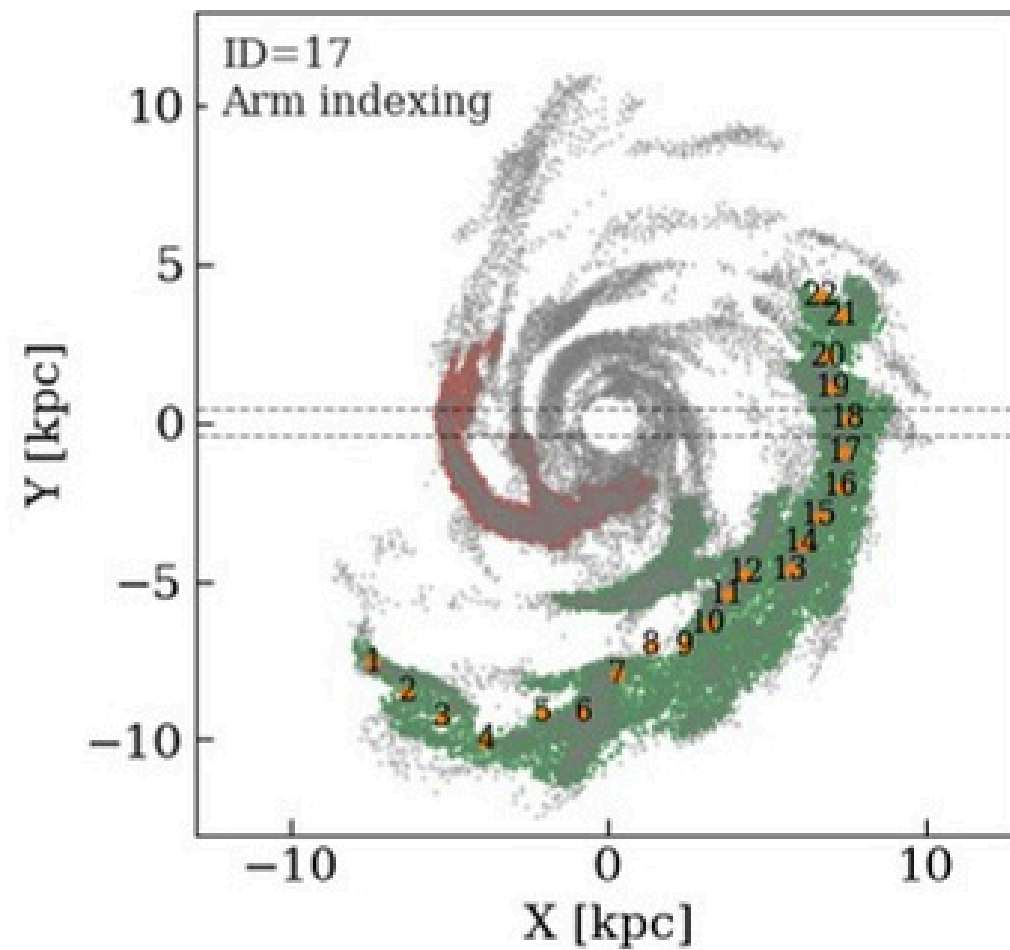
- ☑ Mean Density Profile
- ☑ Cubic Fit for $f(R)$
- ☑ Density Contrast

$$\Delta\rho = \frac{\rho(R, \theta, z)}{f(R)} - 1 \begin{cases} \Delta\rho < 0: \text{Background particles.} \\ \Delta\rho > 0: \text{Overdense regions.} \end{cases}$$



Morphology Measurements

- ☑ Automated Arm Identification
- ☑ Arm-Center Extraction
- ☑ Width Measurement and Fitting

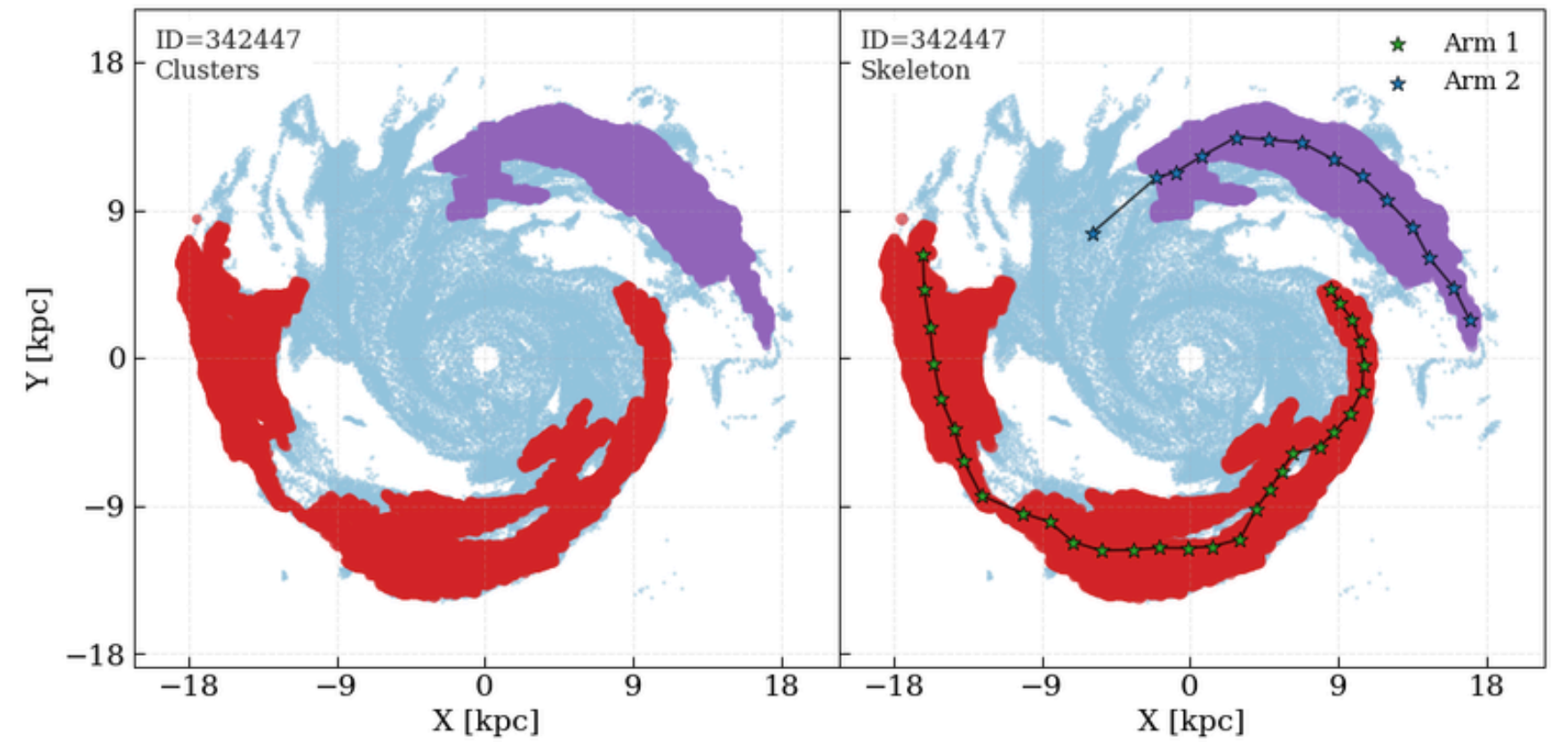
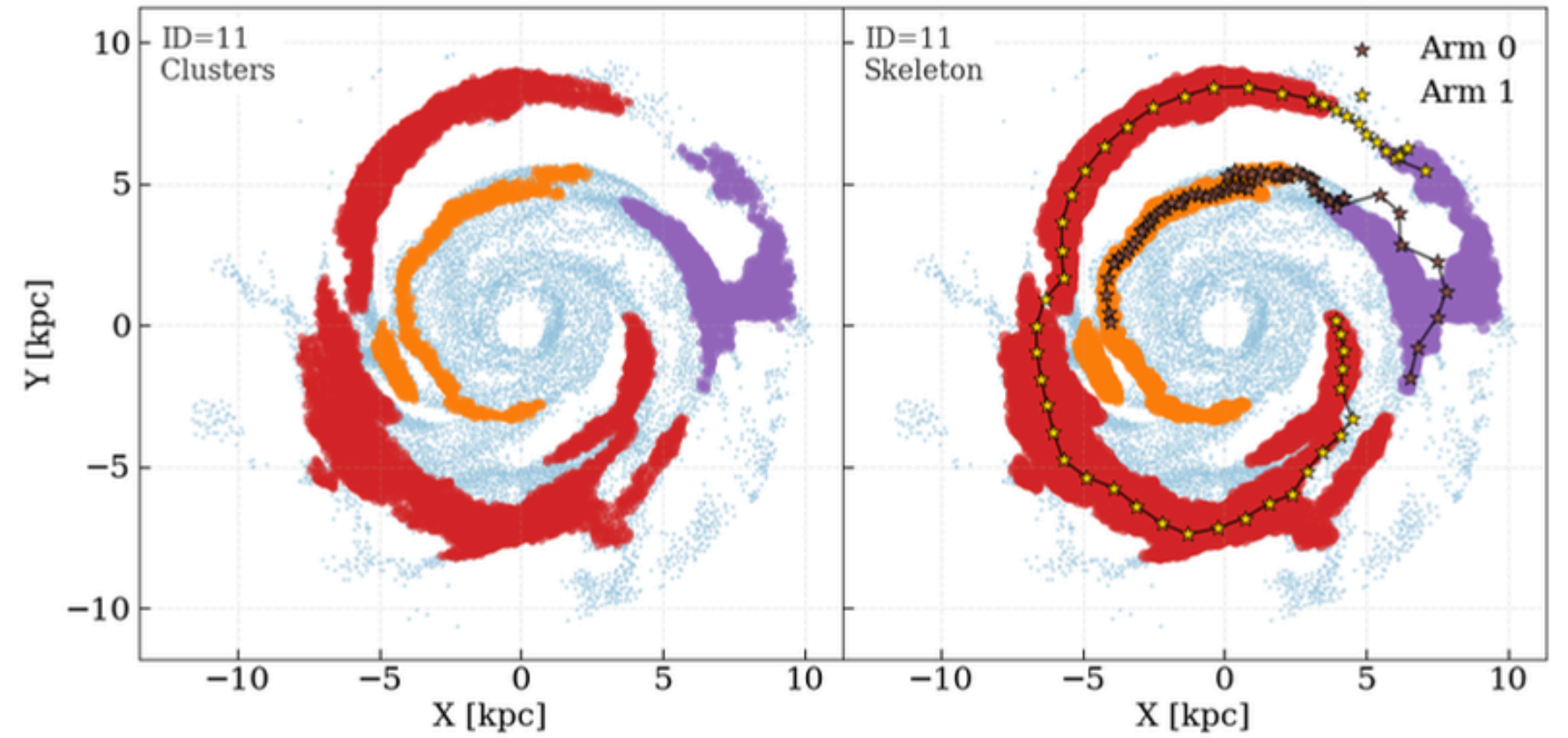
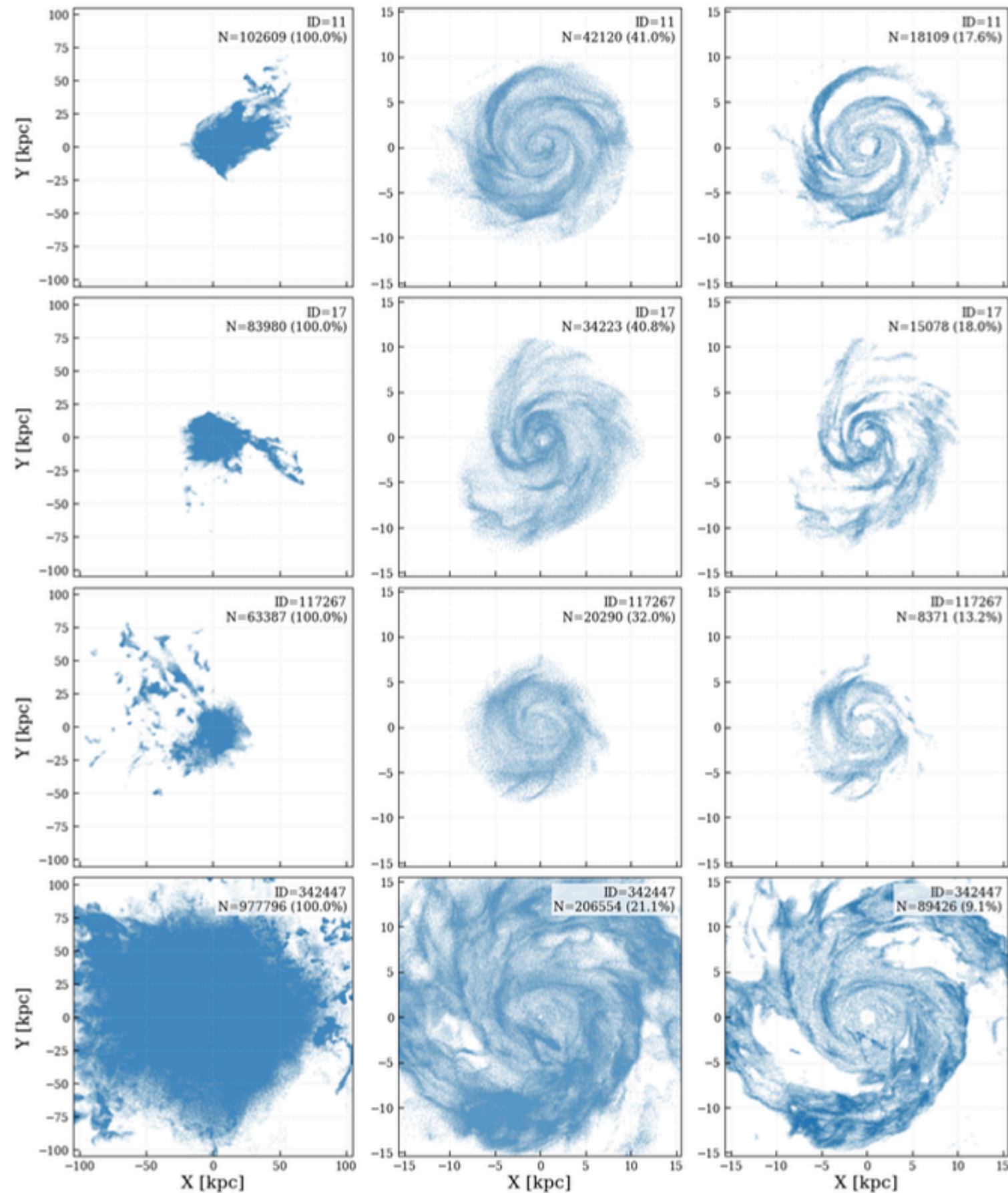


$$G(x) = \frac{A}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} \left[1 + \operatorname{erf}\left(\gamma \frac{x-\mu}{\sqrt{2}\sigma}\right) \right]$$

Asymmetric Gaussian Equation

Results

Method Coverage: ~80%

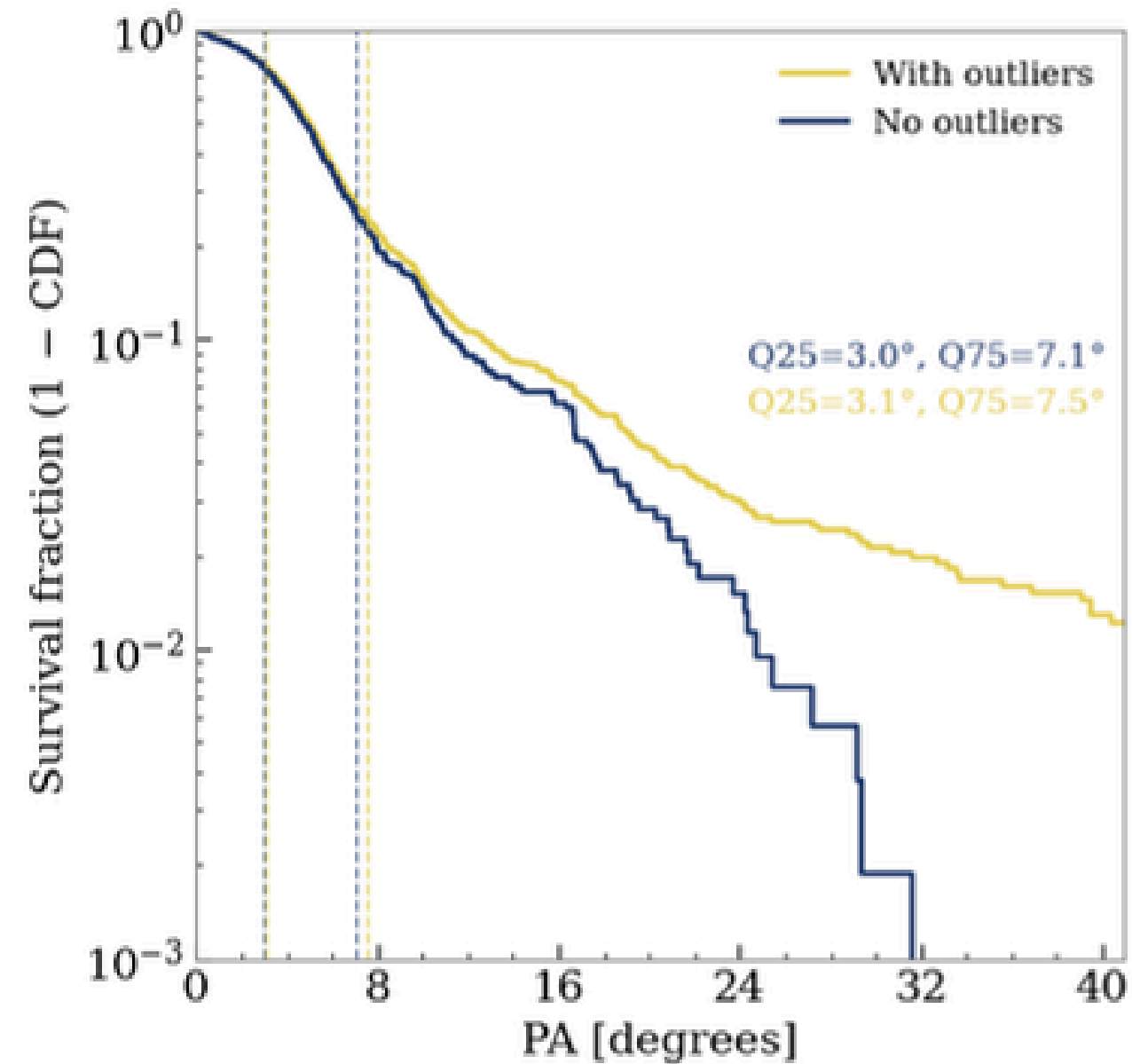


~80% yielded arms: pipeline coverage, not scientific validation.

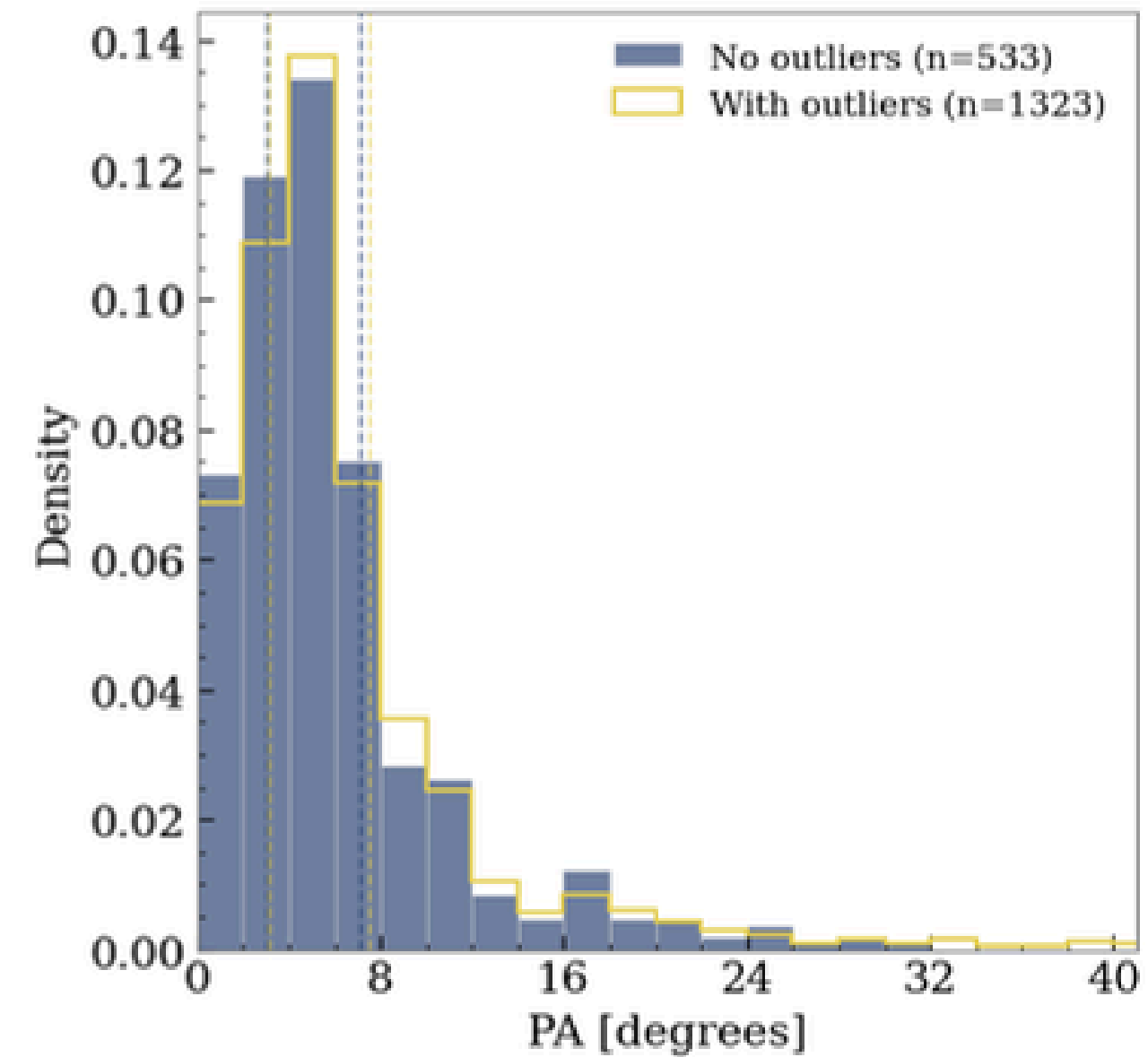
Results

Sample Filters

PA Survival Function



PA Density per Arm



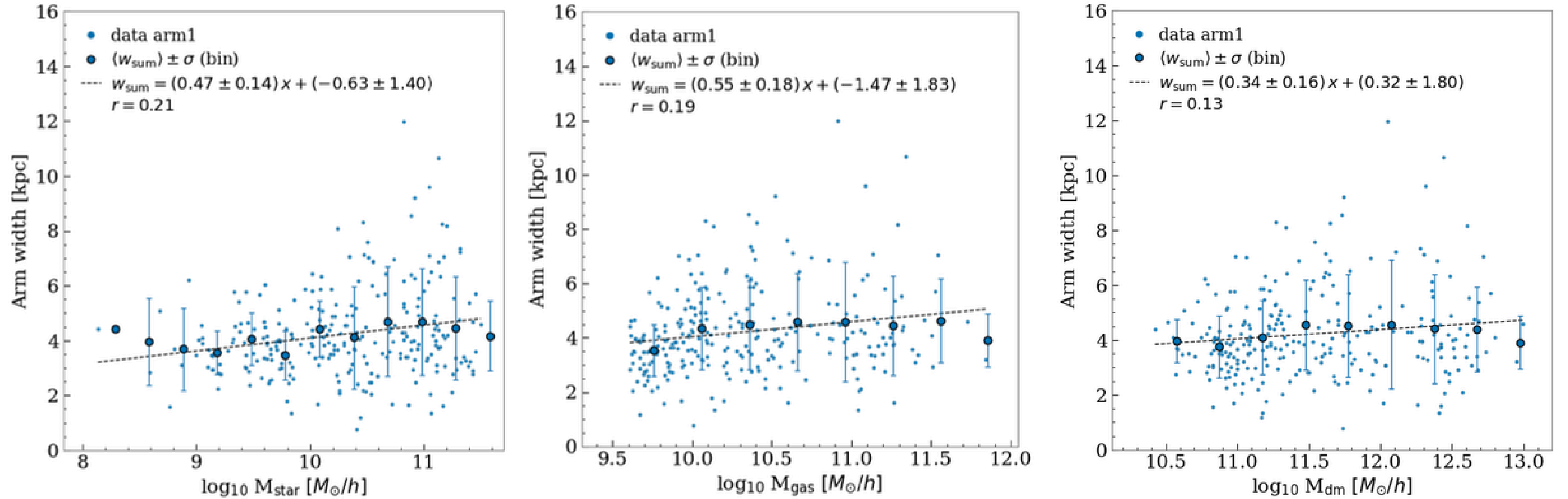
Samples: full $n = 1,323$ (309 disks); clean subset $n = 533$ (267 disks, two PRINCIPAL arms per galaxy)

Mean = 6.85°

Results

Halo Properties:

Width vs. Mass (3 Tracers)

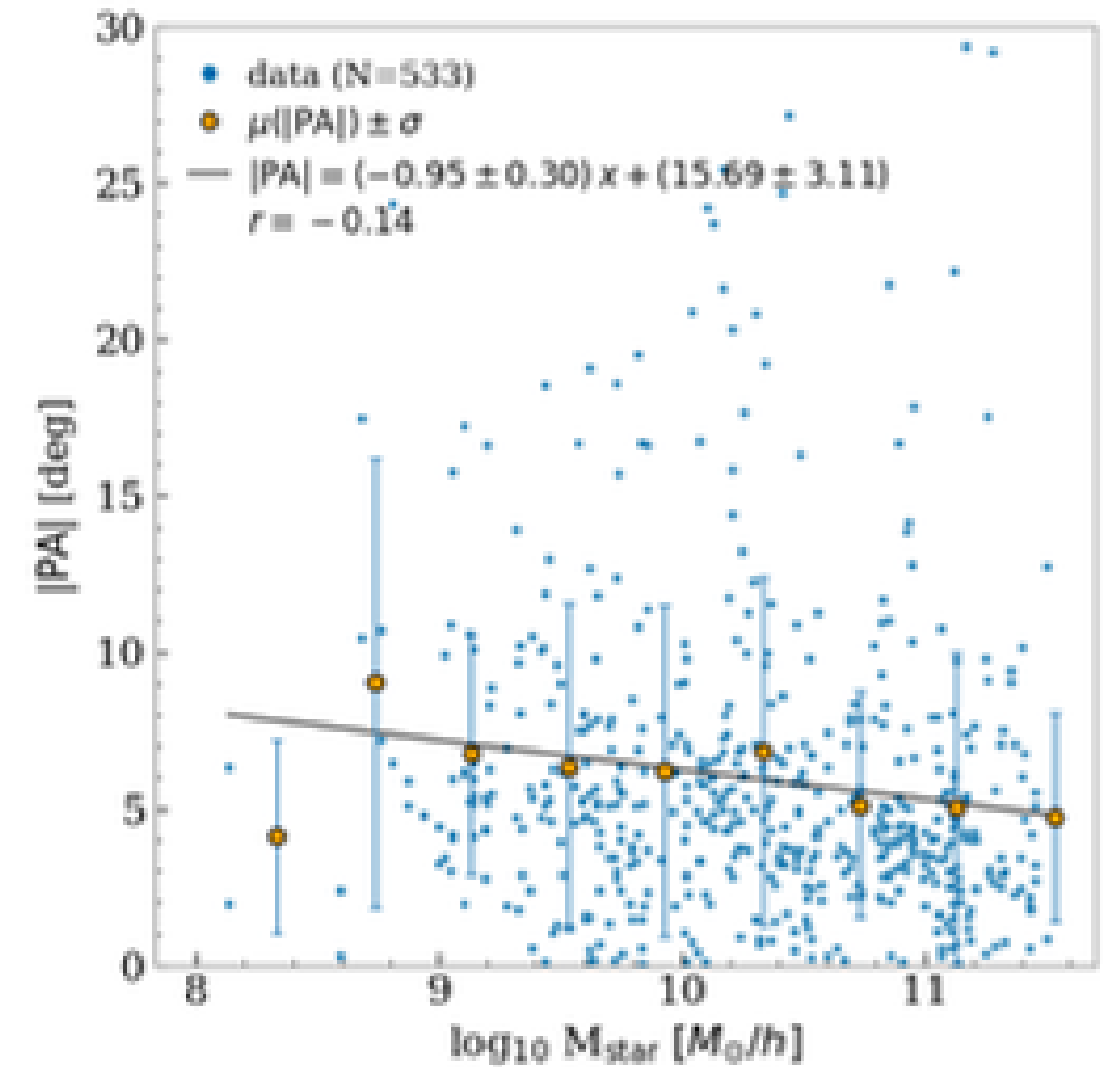
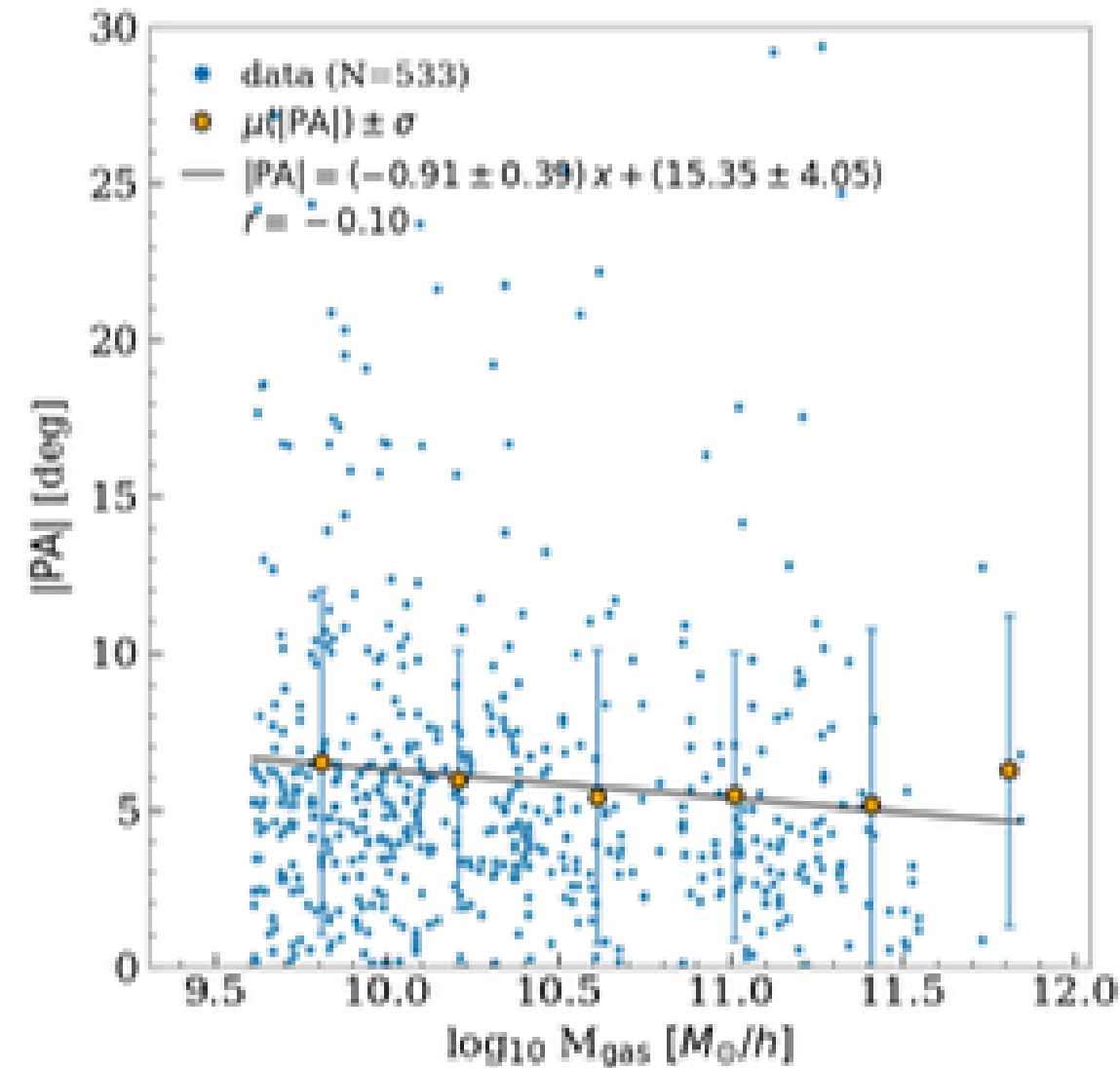
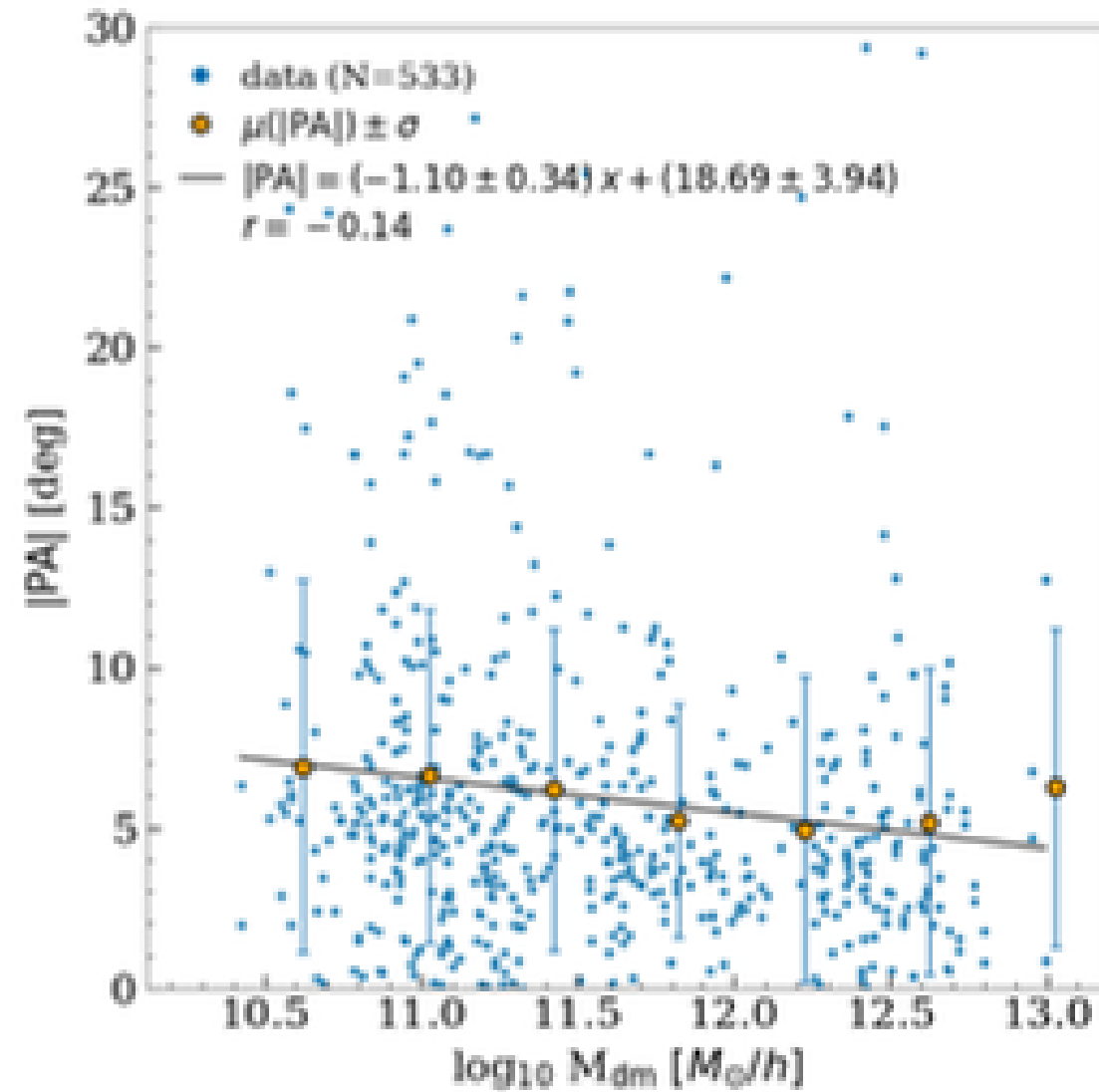


The original estimator suggests wider arms at higher stellar, gas, and dark matter mass.
The trend is tentative ($\leq 1.3\sigma$).

Results

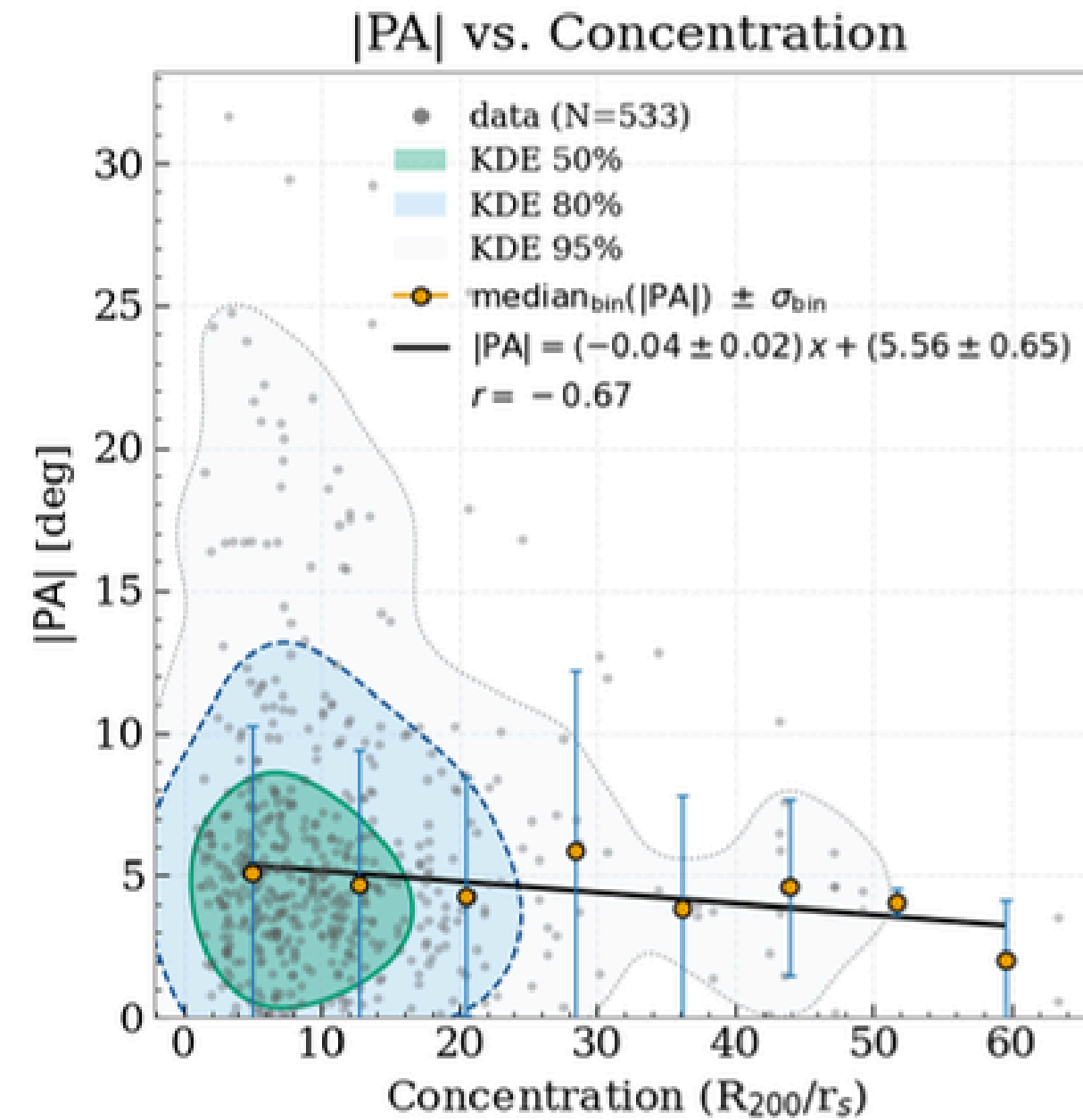
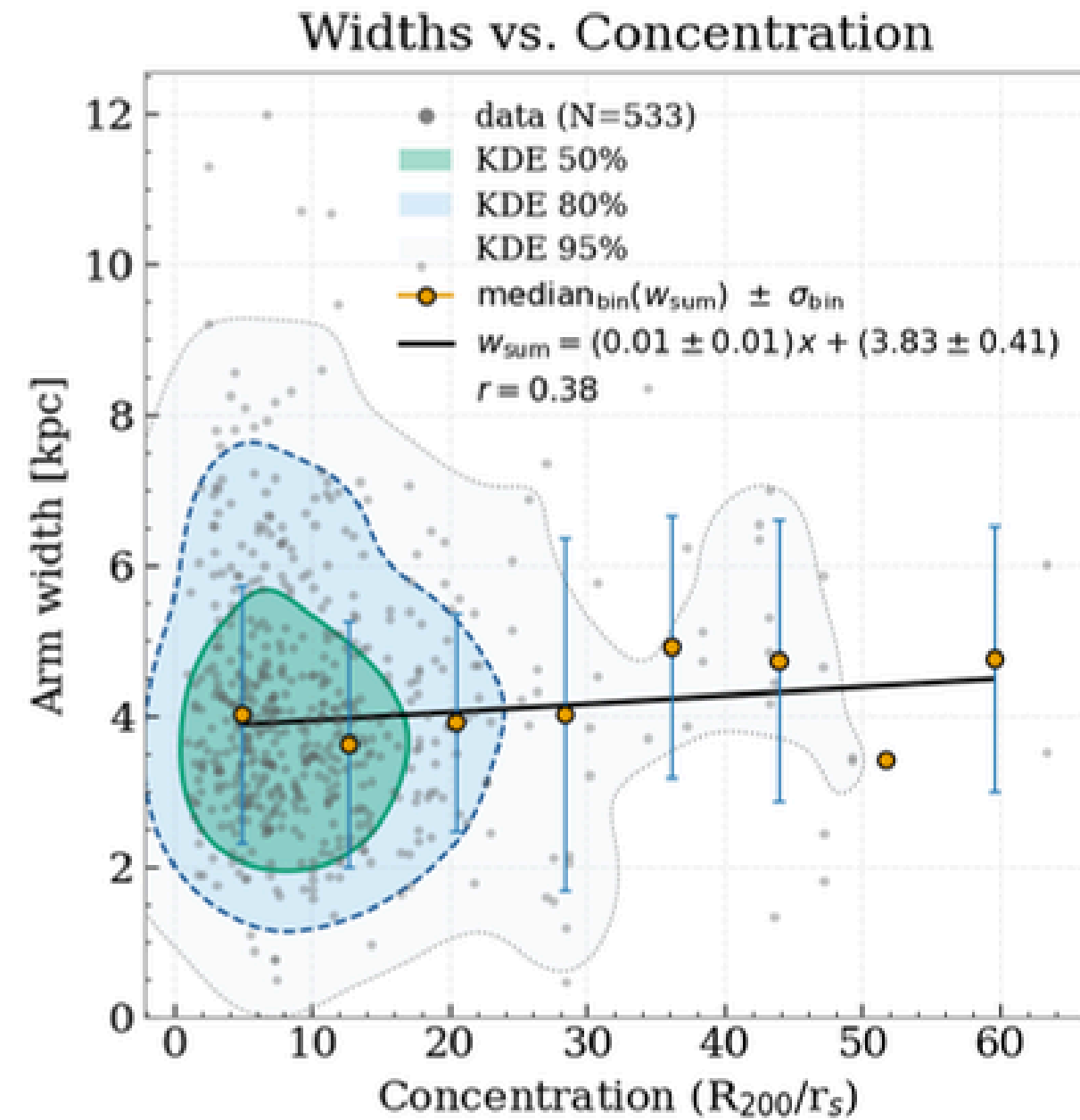
Halo Properties:

Pitch Angle vs. Mass (3 Tracers)

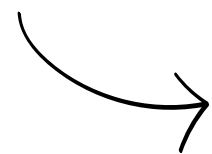


- |PA| decreases mildly with mass; tracer differences are not significant
- More massive systems may host slightly tighter arms, but the evidence is inconclusive.

Results



- Mean arm width is consistent with no concentration trend
- |PA| decreases mildly as concentration increases



Descriptive fit only; broader evidence is tentative and non-causal.

Takeaways

- We developed an astrophysical and numerical workflow that identifies galaxy disks and automatically extracts coherent spiral arms.
- **Arms were recovered in ~80% of filtered disks (method coverage).**
- With the original estimator, mean arm width increases modestly with halo, gas, and stellar mass. This tentative result needs testing with the revised estimator.
- No systematic amplitude–halo-mass trend was detected (Pearson $r \approx 0.06$ – 0.14).
- A weak pitch-angle–concentration anti-correlation appears in this sample. It does not establish that a stronger central potential causes tighter winding.
- Next: rerun revised widths across the full sample, enlarge the galaxy sample, quantify selection effects, and compare with observations.



Sources

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Pillepich, A., et al. (2019). First results from the TNG50 simulation: the evolution of stellar and gaseous discs across cosmic time. *Monthly Notices of the Royal Astronomical Society*, 490(3), 3196–3233.

Silva-Villa, E., & Cano Gomez, X. (2022). NGC 5236's stars as tracers of arms and arm widths in spiral galaxies. *Monthly Notices of the Royal Astronomical Society: Letters*, 514(1), L22–L26.

Yu, S.-Y., & Ho, L. C. (2020). The statistical properties of spiral arms in nearby disk galaxies. *The Astrophysical Journal*, 900(2), 150.

Thank You

Questions and discussion
